

# **CHRONUS INTERVIEW EXPERIENCE**

## **ONLINE TEST:**

Everyone who registered for the Chronus got a link for attending the online test (no candidates were left out due to CGPA). The online test was conducted in the Hackerearth platform where 10 multiple choice questions (1 points each) were asked which are based on Computer fundamentals (like binary trees) and aptitudes (quantitative) and 3 coding questions were asked. You can switch to any questions during the test.

### **Coding questions:**

1. There is an elevator which goes up by 'a' units and goes down by 'b' units. There are some floors where if you are stuck on that floor it implies you cannot move further. Given a floor, determine if it is possible to reach that floor. If it's possible, return the number of steps required, else return -1. Even negative floors are possible.  
(I got only 2 test cases passed in this questions, other test cases where failed due to TLE) - 15 points
2. Given an array of integers, you can increment the elements by 1. You have to make the maximum number of array elements to the minimum possible value in 'k' steps.  
(Again I got only 2 test cases passed in this questions, other test cases where failed due to TLE) - 15 points
3. Given an array of integers, you imagine them as histogram. You could remove rectangles from that histogram. What is the minimum number of rectangles you could remove to completely demolish the whole histogram?  
(No test cases were passed for this question) - 10 points

10 students where shortlisted for the physical interview held at CUIC

## **INTERVIEW**

### **ROUND 1:**

This round completely consists of 2 DSA questions which went for nearly an hour.

1. Given an array consists of number (integer, fraction, negative, zero), you can delete at most one elements in order to maximize the product  
First I gave a brute force approach, followed by optimizing that. I wrote pseudocode in whitepaper.
2. Given a puzzle, decode that and then code that in whitepaper.  
Sample: SUPER  
      SMART  
      +--\*-

Logic is S is present in both strings at the same position, R is present at both strings but at different positions. So, in first case, + symbol was used and in next case \* symbol was used. If it's not a match, then - symbol is used.

Interviewer asked if I had any questions for him.

## **ROUND 2 :**

This round also went nearly for an hour.

I was asked to give a brief description of the project and the interviewer asked me to draw the tables used in my project. I was asked about the choice of primary key and he asked reasons for that. He also asked why did I make the project using that particular tech stack(I used NodeJS, Express, React and MySQL in my project)

Next he asked me some logical puzzles, like measuring time by burning two wires of equal length. Next he asked me some DSA questions. He asked me how much I rate my DSA skills on a scale of 5.

### **1. Detecting cycle in a linked list**

I gave a brute force approach followed by an optimal approach which I could not come up on my own. With the hints given by the interviewer, I developed the approach and wrote the code for that in the white paper.

### **2. Next he gave me a puzzle on arrays, which I was supposed to decode. He gave only one test case, where I thought of some random logic. To make the logic the interviewer had in his mind, I asked for a lot of test cases by telling logics that worked good for the test cases initially provided by the interviewer. The problem was something like this [10, 20, 30, 30, 40] - In case of 60, return true**

The logic was if you consider 20+40, the sum is 60 and the distance between them is 2. So, there are 2 pairs, (20,40) and (30,30) whose sum is 60. Since, the distance between 20 and 40 is also 2 and there are 2 pairs, return true

Interviewer asked if I had any questions for him

## **HR ROUND:**

Firstly, the interviewer asked me to introduce myself. Next he asked where I see myself in three years. Next about the company. He also asked about what I feel about my performance today. The round hardly went for 15 minutes. He also asked about my interest in web development and since I mentioned that I am good at mathematics, he asked about that. Also, regarding why I choose CEG over NITs as I mentioned about my JEE percentile in my resume.

By the end of the day, I got mail that I was shortlisted for the internship. Out of 10 students, 4 of them got the offer for internship.

## **KEY TAKEAWAYS:**

- Solve problems consistently on a daily basis, start from brute force and optimize that (refer to others solution, so, you can have new perspectives)
- Rather than memorizing solutions to the problems, work on different variety of problems to have fresh ideas, that helped me to solve puzzles asked in the interview
- Make the interview lively by explaining your thought process to the interviewer every moment. Remember interviewers are there to HIRE YOU, keep that in mind.
- Think of all possible corner cases while solving a problem
- Ask questions to the interviewer before thinking of the approach. Clearly understand the problem statement before proceeding to solve.
- Add only things that you are confident at in your resume. DO NOT EVER LIE in the resume
- At last, STAY CONFIDENT at any point of time. You can crack any company by practicing DSA and learning new skills (try other things like web development, ML too).

GOOD LUCK!

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